Consolidated Domain Brief for Scenario A

Unified Service Scheduler – Keyloop Technical Assessment

~15-minute read – essential knowledge for designing a solid system

1. Core Business Goal

Turn a customer's service request into a **reliable, executable workshop commitment**. A booking is not just a calendar entry. It is a promise that a specific vehicle will occupy a **Service Bay** and be worked on by a **qualified Technician** for a defined duration, without double-booking or over-promising.

Aftersales generates ~49% of dealership gross profit from only ~12% of revenue. Friction in booking and capacity management is a major reason dealers lose service visits.

2. The Three Mandatory Requirements

Requirement	What it really means
Resource Constrained Booking	Customer (or advisor) supplies Vehicle + Service Type + Dealership + Desired Time. System must treat **bays** and **skilled technicians** as finite resources.
Real-Time Availability Check	Before confirmation, verify that **both** a suitable Service Bay **and** a qualified Technician are free for the **entire** service duration.
Confirmed Appointment Record	Persist a single record linking Customer ↔ Vehicle ↔ Technician ↔ Service Bay ↔ Service Type ↔ Time. This becomes the source of truth for the workshop diary.

3. Key Domain Concepts You Must Model

Service Bay (Workshop Bay / Lift / Stall)

Physical workspace. Finite number per dealership. Not all equal (express/quick-lube, diagnostic, alignment, heavy-vehicle). A car occupies the bay for the full job duration. Industry target ≈ 1.5 bays per technician.

Technician

Has skills/certifications (e.g. ASE electrical, brakes, transmission, general). Has a personal calendar (shifts, already assigned jobs). Must be both free **and** qualified for the service type. Skill mismatch is a major efficiency killer.

Service Type / Product / Work Type

Defines estimated duration (oil change 30–45 min, brake job ~90 min, diagnostic 60–120 min, major repair 4+ h), required skills, preferred or required bay type, and sometimes parts requirements.

Appointment

The persistent record that binds everything together. Once confirmed it feeds the workshop diary, advisor screens, and (in real systems) the DMS.

Capacity / Workshop Diary

Live view of available hours, loaded jobs, and technician allocation. Must prevent over-capacity **before** the customer is promised a slot.

Buffer Time

Never schedule 100% of theoretical capacity. Leave 15–20% buffer for overruns, parts delays, and emergencies. Typical productive target: 85–87.5% technician productivity.

4. Critical Design Principles from Real Systems

• **1. Resource matching is multi-dimensional** — Time slot alone is insufficient. You need simultaneous availability of: bay of the right type, technician with the right skills, and correct duration for that service + vehicle.

• **2. Real-time, not optimistic** — Availability must be checked against live calendars at the moment of booking (exactly what Keyloop's own Service Appointment Booking APIs do).

• **3. Duration-aware slotting** — A 4-hour job cannot be booked into a 2-hour gap. The system must look for a continuous free window of the required length for both bay and technician.

• **4. Skill-based assignment** — Do not give complex diagnostics to the newest tech or oil changes to the master tech. Matching skills improves efficiency and throughput.

• **5. Single source of truth** — Once created, the Appointment record must be the authoritative plan. Changes update the same record so everyone sees consistent status.

• **6. Buffer & resilience** — Build 15–30 min buffers between jobs. Reserve some capacity for walk-ins/emergencies. Track actual vs estimated times so future estimates improve.

5. Typical Happy-Path Flow

1. User selects Dealership + Vehicle + Service Type + preferred time window.

2. System looks up Service Type → duration + required skills + preferred bay type.

3. Search for free continuous slots where a matching bay **and** a matching skilled technician are free for the full duration.

4. Present available slots (or reject + suggest alternatives).

5. On confirmation → create persistent Appointment linking Customer, Vehicle, Technician, and Bay.

6. Optionally trigger confirmation/reminder workflows (out of scope for MVP but good to note).

6. Important Edge Cases & Failure Modes

• Concurrent booking attempts on the same bay/tech (need optimistic locking or transactional check-and-reserve).

• Job overruns that cascade through the day.

• Technician absence or bay unavailable after booking.

• Parts not available (many real systems check this; Scenario A does not require it).

• Walk-ins vs pre-booked capacity tension.

• Different service types needing different bay types or skill levels.

• Multi-day jobs or jobs that span lunch/breaks.

7. What Good Systems Explicitly Do

• Real-time capacity view for service managers.

• Skill + workload aware job allocation.

• Accurate duration per service type (ideally learned from actual shop data).

• Clear separation of "available slot" vs "executable commitment".

• Persistent Appointment as the bridge between customer-facing booking and workshop execution.

8. Scope Guidance for the Assessment

Must implement (core acceptance criteria)

- Resource-constrained request
- Real-time dual availability check (Bay + qualified Technician for full duration)
- Persistent Appointment record

Strongly recommended design considerations

- Clear data model for the four main resources
- Duration-aware continuous-slot search
- Concurrency safety
- Observability (logging of availability checks, booking success/failure)
- Scalable structure (even if you implement only one dealership)

Nice-to-have / future (mention in design doc, do not over-build)

Parts availability, loaner vehicles, multi-channel booking, reminders, upsell during booking, digital check-in.

9. One-Sentence Mental Model

> *"A service appointment is a simultaneous reservation of a physical bay and a skilled human for a continuous block of time whose length is determined by the service type."*

Use this model as the north star for every architectural decision.

Sources: Keyloop Service Hub & Service Appointment Booking materials, Proaxia end-to-end scheduling analysis, practical bay-efficiency guidance (Breakroom), and industry capacity-management practices.